

ROSIA MONTANA, Romania

WMO: 151840

Lat: 46.3175N

Lon: 23.1392E

Elev: 1197

StdP: 87.75

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.9	-11.9	-19.4	0.8	-8.6	-16.6	1.0	-7.9	12.0	-1.7	10.3	-2.5	3.2	40	0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.0	25.1	17.2	23.5	16.6	22.2	16.0	18.3	23.2	17.5	22.1	16.8	21.2	2.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.6	13.7	20.5	15.8	13.0	19.8	15.0	12.3	19.1	57.2	23.5	54.2	22.1	51.8	21.3	21.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.2	7.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.9	-3.9	-1.6	1.0	6.2	11.1	14.9	16.0	17.3	11.8	7.9	3.2	-2.0
	DBStd	8.45	4.74	4.36	4.48	5.74	3.77	3.64	3.41	3.78	4.84	4.99	5.15	4.45
	HDD10.0	1933	431	324	280	139	32	4	2	1	38	103	207	372
	HDD18.3	4246	689	557	537	364	224	111	88	66	202	325	453	630
	CDD10.0	792	0	0	2	25	67	153	187	227	91	37	4	0
	CDD18.3	63	0	0	0	0	1	9	15	33	6	0	0	0
	CDH23.3	158	0	0	0	0	0	13	35	96	14	0	0	0
	CDH26.7	13	0	0	0	0	0	1	1	11	1	0	0	0
Wind	WSAvg	2.9	3.2	3.5	3.3	3.2	2.8	2.5	2.5	2.3	2.8	2.7	3.2	3.3
Precipitation	PrecAvg	809	42	37	48	68	86	101	107	76	72	60	55	60
	PrecMax	1045	88	104	133	129	156	180	179	191	158	142	99	137
	PrecMin	562	13	7	12	9	47	34	30	18	11	3	0	5
	PrecStd	138	21	22	32	27	30	37	41	33	37	37	24	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.4	9.5	14.6	19.4	21.7	24.9	25.7	28.0	25.5	20.1	14.7	9.6
		MCWB	2.1	5.7	7.4	11.5	13.8	18.6	17.5	18.3	15.9	11.5	7.2	4.4
	2%	DB	4.9	7.3	11.4	17.5	19.8	23.1	24.2	25.7	22.1	18.0	12.6	7.2
		MCWB	1.8	4.1	6.0	10.3	13.2	17.1	17.2	17.0	14.9	11.5	7.3	4.0
	5%	DB	3.8	5.4	9.2	15.6	18.2	21.9	22.6	23.9	19.8	16.2	11.3	5.3
		MCWB	1.4	2.9	4.7	9.5	12.6	16.3	16.5	16.6	13.9	10.4	6.5	3.0
	10%	DB	2.6	4.1	7.1	13.6	16.7	20.4	21.0	22.4	18.2	14.5	9.9	3.9
		MCWB	0.6	2.1	3.6	8.5	11.9	15.7	15.8	16.2	13.2	10.1	6.1	1.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	7.2	8.5	12.5	14.9	19.6	18.7	19.2	16.6	13.5	9.6	6.0
		MCDB	5.9	8.7	13.8	18.4	19.6	23.9	23.1	25.3	23.2	16.6	12.6	7.6
	2%	WB	3.4	4.7	6.7	11.2	13.9	18.2	17.9	18.4	15.6	12.2	8.6	4.5
		MCDB	4.3	6.4	10.7	16.5	18.1	22.2	22.6	24.0	21.3	16.0	11.3	6.5
	5%	WB	2.4	3.5	5.2	10.1	13.1	17.1	17.2	17.5	14.5	11.3	7.7	3.4
		MCDB	3.2	4.9	8.9	14.9	17.4	21.1	21.5	22.5	19.2	15.4	10.3	5.0
	10%	WB	0.8	2.5	3.9	8.9	12.2	16.1	16.5	16.7	13.6	10.4	6.8	2.4
		MCDB	2.0	3.8	6.6	13.0	16.1	19.9	20.5	21.4	17.9	14.1	9.0	4.0
Mean Daily Temperature Range	5% DB	MDBR	4.7	5.0	6.3	7.3	6.8	7.1	7.2	7.0	6.1	5.7	4.6	4.4
		MCDBR	4.7	5.9	8.5	8.7	8.5	8.6	8.2	8.0	7.5	7.1	5.7	5.2
		MCWBR	3.4	4.0	5.4	4.8	5.0	4.9	4.6	4.3	4.0	4.0	3.6	3.5
	5% WB	MCDBR	4.3	5.5	8.1	8.3	7.9	8.5	7.5	7.8	7.1	6.8	5.2	4.9
		MCWBR	3.5	4.1	5.3	4.8	5.0	5.1	4.7	4.5	4.1	4.2	3.5	3.8
Clear-Sky Solar Irradiance	taub	0.283	0.309	0.348	0.395	0.383	0.377	0.393	0.399	0.364	0.333	0.309	0.285	
	taud	2.379	2.331	2.294	2.193	2.279	2.320	2.276	2.258	2.336	2.420	2.440	2.416	
	Ebn at Noon	818	862	874	859	883	888	867	846	846	824	780	775	
	Edh at Noon	72	92	112	136	129	124	128	126	106	84	68	63	
All-Sky Solar Radiation	RadAvg	1.23	2.02	3.16	4.33	5.21	5.78	5.66	5.22	3.71	2.56	1.48	0.98	
	RadStd	0.17	0.26	0.46	0.46	0.45	0.49	0.50	0.50	0.41	0.30	0.20	0.13	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (70 neighbors)	+0.71	N/A	N/A	N/A	+1.00	N/A	N/A	-125	-189	+121	+60

Nomenclature: See separate page